Tong Chen

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Research Interests

Machine learning for neuroimaging and neurodegenerative disease — particularly Alzheimer's disease and Lewy body dementia (spanning MCI to dementia) — with an emphasis on graph-based representations of brain structure and connectivity, disease-progression modeling, and interpretable, multi-cohort clinical analysis.

Education

Ph.D. in Computer Science Aug 2023 – Present (expected Dec 2026)
The University of Texas at Arlington *Advisor: Dr. Dajiang Zhu*

M.S. in Computer Science Jan 2021 – Dec 2022
Kennesaw State University

B.S. in Computer Science Aug 2012 – May 2018
Kennesaw State University

Honors & Awards

- John Stephen Schuchman Memorial Outstanding PhD Student Scholarship Award, Dept. of Computer Science & Engineering, UT Arlington, 2026

Publications

(**Bold** = author; first / co-first authorship noted.)

Journal Articles

1. Lu Zhang, Junqi Qu, Haotian Ma, **Tong Chen**, Tianming Liu, and Dajiang Zhu. "Exploring Alzheimer's Disease: A Comprehensive Brain Connectome-Based Survey." *Psychoradiology*, kkad033, 2024.

Refereed Conference Proceedings

1. **Tong Chen**, Minheng Chen, Jing Zhang, Chao Cao, Li Su, Tianming Liu, and Dajiang Zhu. "HiLoGraph: Hierarchical-Longitudinal Brain Network Representation Learning." *Intl. Conf. on Medical Image Computing and Computer Assisted Intervention (MICCAI)*, 2026. **(First author; early accept, top 9%)**
2. **Tong Chen**, Minheng Chen, Jing Zhang, Yan Zhuang, Chao Cao, Xiaowei Yu, Yanjun Lyu, Lu Zhang, Li Su, Tianming Liu, and Dajiang Zhu. "A Unified Continuous Staging Framework for Alzheimer's Disease and Lewy Body Dementia via Hierarchical Anatomical Features." *Intl. Conf. on Medical Image Computing and Computer Assisted Intervention (MICCAI)*, Daejeon, South Korea, 2025. **(First author; early accept, top 9%)**
3. Chao Cao*, **Tong Chen***, Nan Zhao*, Minheng Chen, Minghao Qu, Zhiyuan Zhang, Xin Shi, Xiang Li, Tianming Liu, and Dajiang Zhu. "Gyral-Sulcal-Net: An Integrated Network Representation of Brain Folding Patterns." *IEEE Intl. Symposium on Biomedical Imaging (ISBI)*, 2026. **(Co-first author)**
4. Chao Cao, Xiaowei Yu, Lu Zhang, **Tong Chen**, Yanjun Lyu, Tianming Liu, and Dajiang Zhu. "Enhancing Group-Wise Consistency in 3-Hinge Gyrus Matching via Anatomical Embedding and

Structural Connectivity Optimization." *IEEE Intl. Symposium on Biomedical Imaging (ISBI)*, Athens, Greece, 2024.

5. **Tong Chen**, Jiho Noh, Luke Cranfill, John Morris, and Junggab Son. "Improving Fashion Attribute Classification Accuracy with Limited Labeled Data Using Transfer Learning." *21st IEEE Intl. Conf. on Machine Learning and Applications (ICMLA)*, Nassau, The Bahamas, 2022. Oral presentation (acceptance rate 32.4%). **(First author)**

Conference Abstracts

1. **Tong Chen**, Minheng Chen, Yan Zhuang, Jing Zhang, Lu Zhang, Tianming Liu, Andrew M. Blamire, John-Paul Taylor, Li Su, John T. O'Brien, and Dajiang Zhu. "Representing the Cognitive Impairment Continuum of Alzheimer's Disease and Lewy Body Dementia with a Novel Finer-Scale Cortical Representation via Disease Embedding Tree." Alzheimer's Association International Conference (AAIC); abstract published in *Alzheimer's & Dementia*, 21, e102179, 2025. **(First author)**

Selected Presentations

- Presented first-author paper, MICCAI, Daejeon, South Korea, 2025.
- Oral presentation, ICMLA, Nassau, The Bahamas, 2022.

Research Experience

Graduate Research Assistant — The University of Texas at Arlington Aug 2023 – Present
Advisor: Dr. Dajiang Zhu

- Developed graph neural network methods for multi-class dementia classification and continuous disease-progression modeling using multi-cohort neuroimaging data (ADNI, NACC, and in-house LBD datasets).
- Authored first-author papers at MICCAI 2025 and MICCAI 2026 (both early accept) and a first-author AAIC 2025 abstract, and co-authored work across MICCAI, ISBI, and other venues.
- Preprocessed and analyzed multimodal neuroimaging (T1-weighted, diffusion, and functional MRI) using FreeSurfer, FSL, SPM12, and CAT12.
- Contributed to drafting sections of federal research proposals.

Research Assistant — University of Nevada, Las Vegas Mar 2023 – May 2023
Advisor: Dr. Junggab Son

- Researched interpretability methods for deep learning and generative models.

Graduate Research Assistant — Kennesaw State University Jan 2021 – Dec 2022
Advisors: Dr. Junggab Son; Dr. Jiho Noh

- Researched computer vision, natural language processing, generative models, and security of deep learning algorithms; published at ICMLA 2022.

Technical Skills

Programming: Python, MATLAB, LaTeX
ML / DL Frameworks: PyTorch, PyTorch Geometric, scikit-learn
Neuroimaging Tools: FreeSurfer, FSL, SPM12, CAT12
Methods: Graph Neural Networks, Transformers, CNNs, Representation Learning, Domain Adaptation

Academic Service

- Program Committee Reviewer, Intl. Conf. on PErvasive Technologies Related to Assistive Environments (PETRA), 2026.

References

- Dr. Dajiang Zhu (Ph.D. advisor), Dept. of Computer Science & Engineering, University of Texas at Arlington.
- Dr. Tianming Liu, School of Computing, University of Georgia.
- Dr. Li Su, University of Cambridge / University of Sheffield; Dr. John T. O'Brien, Dept. of Psychiatry, University of Cambridge.

Full contact details available upon request.